

ASSIGNMENT COVER SHEET

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STUDENT ID(s): (DO NOT INCLUDE YOUR NAME)	2951140, 3064876, 3051444, 3030456			
MODULE TITLE & CODE: <i>E.g: Epilepsy - 17623</i>	Data Analytics & Statistical Machine Learning (37237 & 37911)			
MODULE LEAD: <i>Insert Name(s)</i>	Dr. Mamunur Rashid			
ASSIGNMENT TITLE: <i>E.g. Question answered from brief.</i>	Group Work with Individual Essay Submission			
Contribution of each member (%). Should add up to 100%	Student 1	Student 2	Student 3	Student 4
	25%	25%	25%	25%
WORD COUNT:				
DEADLINE DATE: <i>Original deadline date</i>	9th May, 2026	DATE SUBMITTED:	08th May, 2026	
EXTENSION DATE:	13th May, 2026			

University of Birmingham Assessment and Feedback

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Data Analysis and Statistical Machine Learning– Group 5 Analysis Report

Abstract:

Neuroblastoma accounts for ~15% of pediatric cancer-related deaths worldwide. The goal of this analysis was to identify a lean molecular signature for predicting death from disease using RNA-seq and microarray data. Our feature selection pipeline produced a 20-feature signature per platform. XGBoost was found to be the best classifier from those we tested, achieving high ROC-AUC (microarray 0.919, RNA-seq 0.937). We also explore the ability of the model to identify high-risk patients missed by INSS staging and the relevance of the selected features. Despite RNA-seq revealing transcriptional complexity invisible to microarrays, the performance advantage was modest, and the microarray model achieved superior recall. Microarray remains more pragmatic for immediate deployment, while RNA-seq retains value for biomarker discovery.

1. Introduction

Microarrays have been widely used for decades as the primary platform for gene expression analysis, and many methods and models have been developed based on them. With the emergence of RNA-seq as the dominant technique for measuring gene expression, the question of how, and whether, this knowledge can be transferred between platforms has become increasingly important (Su et al., 2014). Additionally, microarrays remain less expensive than RNA-seq, making it valuable to investigate whether RNA-seq provides additional clinically relevant information for prediction tasks.

The data for this project were originally generated as part of a benchmarking effort by the SEQC/MAQC-III Consortium to compare microarray and RNA-seq technologies and evaluate whether models based on RNA-seq can achieve similar predictive success in clinical use cases as microarrays (SEQC/MAQC-III Consortium, 2014; Zhang et al., 2015).

The cohort comprises 498 neuroblastoma patients. Neuroblastoma is a pediatric cancer that develops from tissues of the sympathetic nervous system. It affects boys slightly more often than girls and accounts for approximately 15% of pediatric cancer-related deaths worldwide (Nong et al., 2025). One of the most well-established indicators of treatment outcome and survival is the presence or absence of amplification of the proto-oncogene *MYCN* (Huang and Weiss, 2013). The dataset contains information on multiple clinical features, including age and sex, as well as longitudinal data on disease outcomes.

The main goal of this analysis is to use both the microarray and RNA-seq data to train separate models to predict death from disease within this cohort. The aim is to identify a lean signature capable of confidently distinguishing between the two outcomes while also comparing predictive performance between microarray- and RNA-seq-based models.

Previous studies have explored more complex signatures (Zhang et al., 2015), as well as highly reduced signatures focused on specific pathways (Chen et al., 2022; Liu et al., 2024), alongside several additional approaches, including deep learning models (Feng et al., 2021). This analysis focuses on an intermediate number of features in an attempt to capture complex biological relationships while maintaining a model that remains as clinically deployable as possible.

2. Methods

We used gene expression profiles from 498 primary neuroblastomas using RNA-Seq and microarrays. The neuroblastoma cohort was randomly divided into training and validation sets, and predictive models on clinical endpoints were generated and evaluated.

2.1. Data Cleaning and Alignment

Extra dots in the headers and prefixes (e.g. FactorValue..) where necessary have been stripped from the metadata to make it programmatically accessible. While working towards minimizing biological inconsistencies across datasets and strictly matching the patient ID across the datasets to perform a thorough, complete-case analysis, expression matrices were limited to numeric data with a null-value drop.

2.2 Unsupervised Learning

Standardized expression matrices were then subjected to PCA to obtain linearly independent expression components from high-dimensional expression data. In this step, unsupervised visualization of the global data set structure by clustering was used to find clinical groupings or batch effects before any supervised labels.

Table 1: Parameters used for Unsupervised (PCA) learning

PCA Parameters	
Parameter / Setting	Value / Description
Variance Threshold	threshold=0.0
Feature Selection	TOP_N = 2000
Feature Scaling	StandardScaler()
PCA Components	n_components=2
SVD Solver	svd_solver="full"
Random State	42

2.3 Feature Selection

The feature selection pipeline started with the use of a variance-based filter, which retains only the top 25% features and removes low-information features that are more likely to introduce noise than a meaningful biological signal. Next, the data was scaled using `StandardScaler` fitted on the training data. Pearson correlation-based filtering was performed to deal with the redundancy. When the correlation coefficient $|r| > 0.90$, the algorithm returned the feature with the highest mean absolute correlation (with the "death.from.disease" target) to reduce redundancy and retain the discriminating power. Basically, it keeps the feature that is the most representative of that cluster of correlated features. A feature count size sweep was performed to determine the optimum number of features based on 5-fold CV ROC-AUC. Finally, the optimum 'lean' signature of genes was determined using Recursive Feature Elimination (RFE).

Table 2: Parameters used for feature selection

Recursive Feature Elimination (RFE) Parameters	
Component	Details
Estimator	Random Forest Classifier with 100 estimators
Selection Target	10 features per dataset
Step Size	1 features removed per iteration
Rationale	Random Forests capture non-linear relationships and feature interactions. RFE iteratively prunes the least important features, identifying the most robust "gene signature" for mortality prediction.

2.4 Supervised Learning

Six architectures (Elastic Net, Random Forest, PLS-DA, Logistic Regression, NSC, and XGBoost) were compared to choose the optimal model, with distribution difference addressing using platform-specific regularization and class weighting.

Table 3: Table showing the parameters across different models used to select the optimal model

Supervised Learning Model Parameters					
Parameter / Setting	Elastic Net	Random Forest	PLS-DA	Logistic Regression (L2)	Nearest Centroid (NSC)
Penalty	elasticnet	Bagging (Ensemble)	Latent Projections	L2 (Ridge)	Centroid Shrinkage
Regularization	C=0.5	Tree Constraints	5 Components	C=0.001 (RNA) / 0.01 (Micro)	Shrink threshold = 0.1
Specific Setting	l1_ratio=0.5	n_estimators=80, max_features='sqrt'	N/A	N/A	0.2 (RNA) / 0.1 (Micro)
Class Weighting	balanced	balanced	N/A	balanced	N/A
Solver	saga	Decision Tree Ensemble	Iterative Least Squares	liblinear	Euclidean Distance
RNA Threshold	0.706	0.5	0.278	0.55	0.3
Micro Threshold	0.835	0.5	0.544	0.55	0.3

2.5 Optimal Model- XGBoost

The same "lean" feature set identified for mortality prediction was also utilized to predict additional clinical endpoints (high_risk, inss_stage, and progression) using specialized XGBoost objectives (binary: logistic and multi:softprob).

To test whether RNA-seq's transcript-level resolution provides a measurable advantage over Microarray's gene-level probe intensities, the selected RNA-seq features were cross-referenced against the microarray probes.

Table 4: Highlighting the parameters used for training and prediction

XGBoost Parameters			
Parameter	RNA-seq Value	Microarray Value	Rationale
n_estimators	338	363	Specifies the number of trees; both datasets took more than 300 trees to settle to a consistent prediction.
learning_rate	0.151	0.118	Regulates the step size so as not to overfit; the RNA-seq model used a slightly larger learning rate.
max_depth	4	2	Has limited number of trees in each tree. Hold-Outs on the Microarray data had superior generalization using shallower trees.
gamma (γ)	4.177	2.427	The pseudo-regularization parameter will always force the split if the amount of reduction in the losses is at or greater than this.

min_child_weight	7	10	Avoids creating nodes that represent insufficient samples, especially for small clinical cohorts.
subsample	0.97	0.7	The proportion of samples taken for each tree. Both RNA-seq and Microarray used almost the whole data set, with a subset being used for diversity in Microarray.
colsample_bytree	0.602	0.614	Randomly selecting part of the features (genes) per tree helps to 'de-correlate' the individual estimators.
colsample_bylevel	0.908	0.637	Further adjustment of the model's robustness through subsampling of the lowest levels of the tree.
reg_alpha (L1)	1	0.1	L1 regularization of weights to promote sparsity of the feature importance scores.
reg_lambda (L2)	1	5	Weight regularization: L2. The Microarray model needed much higher L2 penalties to be sufficiently stable.
Optimal Threshold	0.74	0.43	The learning rule for selection of the decision boundary is, mathematically speaking the maximisation of the Matthews Correlation Coefficient (MCC).
Max MCC	0.712	0.664	The best performance was obtained on the training set as out-of-fold cross-validation results.

3. Results:

3.1. Unsupervised Machine Learning

PCA on the top 2000 variable genes showed no clear clustering of death outcome or INSS stage (Figure 1) with significant overlap between groups, suggesting that gene expression patterns don't naturally separate into distinct clinical groups.

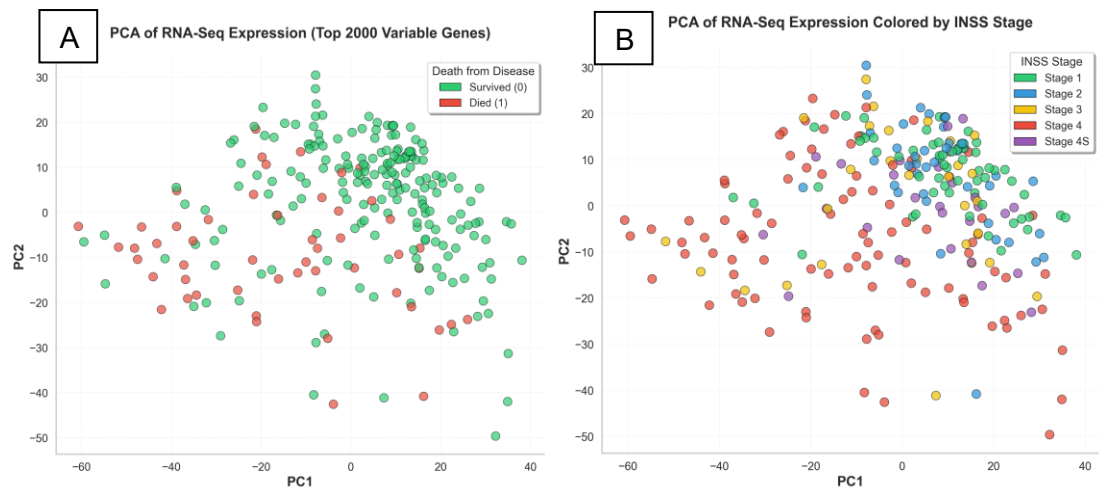


Figure 1: PCA of RNA-Seq top 200 variable genes grouped by **A:** “Death from disease” and **B:** “INSS Stage”

3.2. Feature selection result

The variance filter (top 25%) selected 5,787 from 23,146 RNA-seq features and 10,823 from 44,708 microarray features. A Pearson's correlation filter ($r > 0.9$) further reduced these to 4,833 RNA-seq features and 6,939 microarray features. Finally, Recursive Feature Elimination with a Random Forest estimator selected the top 20 features (Figure 2A & 2B).

A feature count size sweep ($n = 5$ to 60) confirmed that performance plateaued around $n \approx 20$ (AUC = 0.944) and $n \approx 12$ (AUC = 0.928) for the microarray and RNA-seq models, respectively. (Figure 2C & 2D) Paired t-tests with Cohen's d effect size confirmed no statistically significant gain beyond $n=20$. (Table 5)

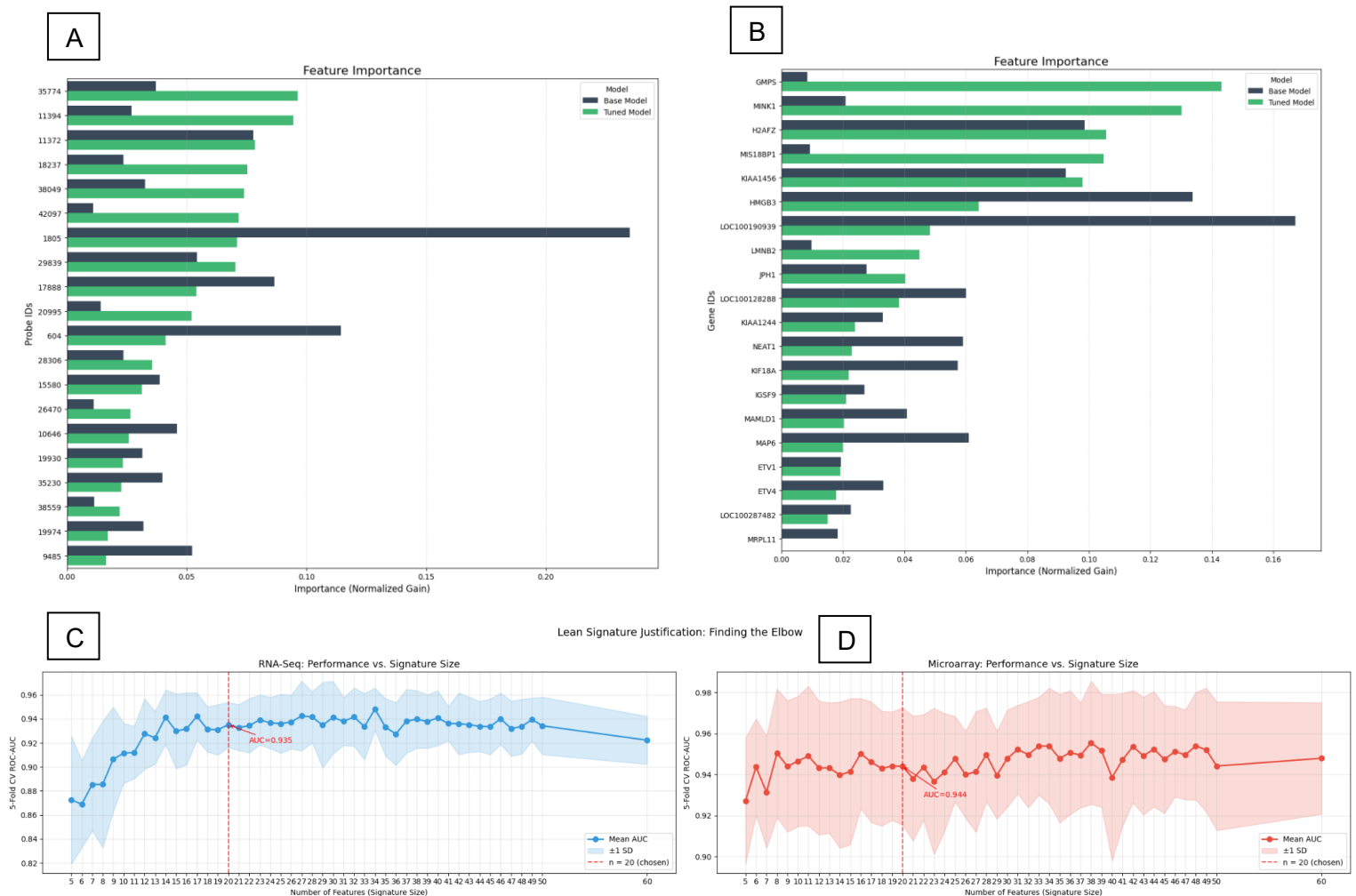


Figure 2.A & Figure 2.B: Top 20 Microarray and RNA-seq features ranked by XGBoost importance.
Figure 2.C & Figure 2.D: Feature count size sweep.

Table 5. Paired t-test and effect size to decide feature count

Platform	Features Compared (X vs Y)	Mean Score (X)	Mean Score (Y)	Mean Diff (X-Y)	t-statistic	p-value	Cohen's d (Effect Size)	Clinical Inference
RNA-Seq	30 vs 20	0.9412	0.9351	0.0061	-0.7034	0.5206	0.3517 (Small)	Not significant; prefer n=20.
Microarray	30 vs 20	0.9478	0.9440	0.0038	-0.5755	0.5958	0.2878 (Small)	Not significant; prefer n=20.
RNA-Seq	20 vs 10	0.9351	0.9114	0.0236	-2.3204	0.0811	1.1602 (Large)	Significant, but large effect size makes test underpowered and inconclusive.
Microarray	20 vs 10	0.9440	0.9465	-0.0025	0.1883	0.8598	-0.0942 (Negligible)	Not significant, but visual check prefers n=20.

3.3. Supervised Machine Learning

XGBoost consistently ranked highest across F1-score, MCC, and ROC-AUC on both platforms. Microarrays' lower dimensionality and technical characteristics may exhibit simpler decision boundaries. Despite differences between data types, XGBoost showed robust generalizability by maintaining strong performance across both platforms. Therefore, XGBoost was chosen as the final model for downstream evaluation, given its consistent performance in key metrics and ability to handle non-linear patterns (Tables 6 & 7).

Table 6. RNA-seq Model Performance

Model	Accuracy	Balanced Accuracy	Specificity	F1	MCC	ROC-AUC	PR-AUC
XGBoost	0.91	0.82	0.97	0.76	0.71	0.94	0.84
Random Forest	0.86	0.85	0.86	0.71	0.63	0.93	0.80
Elastic Net	0.87	0.81	0.91	0.69	0.61	0.91	0.71
Linear SVC	0.87	0.76	0.95	0.64	0.58	0.90	0.68
PLS-DA	0.79	0.81	0.77	0.62	0.52	0.87	0.62
Logistic	0.80	0.85	0.77	0.65	0.58	0.90	0.70
NSC	0.83	0.85	0.81	0.68	0.60	0.91	0.70

Table 7. Microarray Model Performance

Model	Accuracy	Balanced Accuracy	Specificity	F1	MCC	ROC-AUC	PR-AUC
XGBoost	0.87	0.87	0.86	0.73	0.66	0.92	0.75
Random Forest	0.86	0.83	0.87	0.69	0.61	0.92	0.75
Elastic Net	0.89	0.78	0.96	0.69	0.63	0.90	0.72
Linear SVC	0.87	0.71	0.99	0.58	0.56	0.91	0.77
PLS-DA	0.86	0.72	0.95	0.58	0.51	0.86	0.64
Logistic	0.82	0.87	0.79	0.69	0.62	0.91	0.73
NSC	0.82	0.83	0.82	0.66	0.57	0.90	0.67

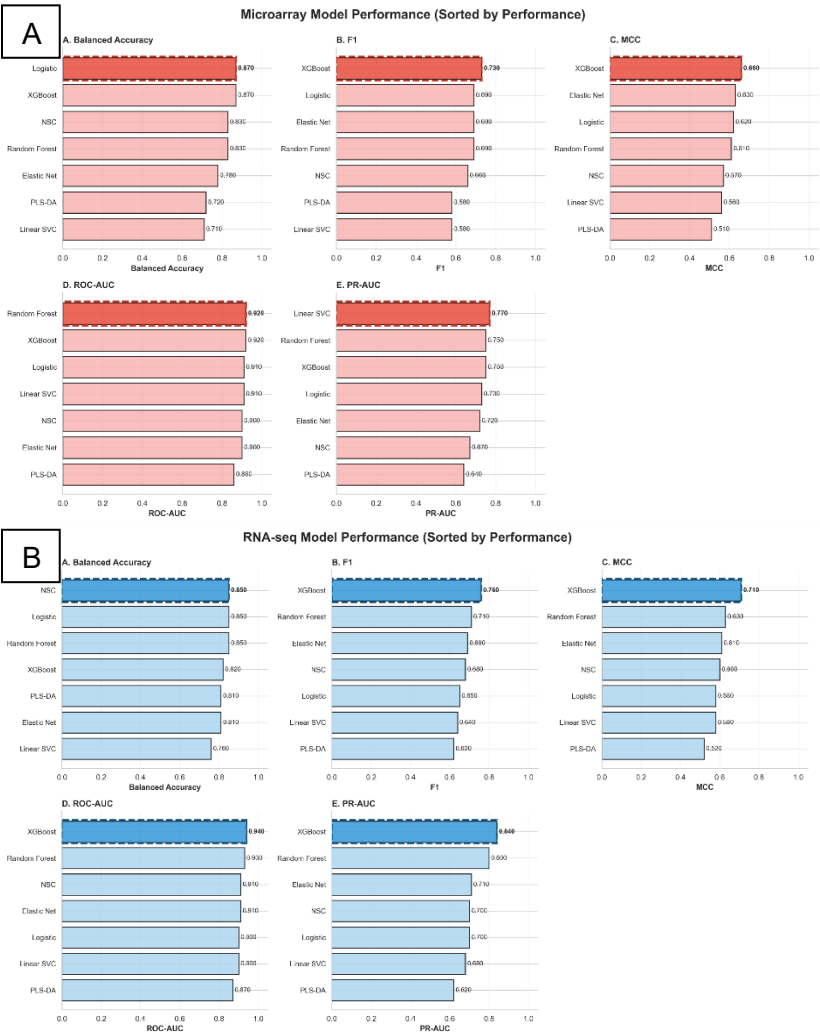


Figure 3.A: Performance of different microarray models 3.B: Performance of different RNA seq models

Table 8. Predicted clinical endpoints for the XGBoost model trained on RNA-Seq and Microarray data

Clinical Endpoint	Class	RNA-Seq (%)	Microarray (%)
Death from disease	0	213 (85.5%)	172 (69.1%)
	1	36 (14.5%)	77 (30.9%)
High risk	0	151 (60.6%)	150 (60.2%)
	1	98 (39.4%)	99 (39.8%)
INSS stage	1	125 (50.2%)	95 (38.2%)
	2	3 (1.2%)	16 (6.4%)
	4	121 (48.6%)	123 (49.4%)
	4S	—	15 (6.0%)
Progression	0	140 (56.2%)	134 (53.8%)
	1	109 (43.8%)	115 (46.2%)

3.4. Model robustness evaluation

We evaluated model generalization and detected overfitting based on ROC-AUC and gaps between training and CV-AUC (gap <0.06: acceptable, >0.06: concerning overfitting). Overall, regularized XGBoost showed acceptable overfitting and achieved the best balance between high predictive performance and generalization, justifying its selection as the final model (Tables 9 & 10).

Table 9. RNA-Seq Models - Overfitting Analysis

Model	Train AUC	CV AUC	AUC Gap	Assessment
XGBoost	0.9800	0.9384	0.0416	Mild overfitting, acceptable
Random Forest	0.9755	0.9267	0.0488	Mild overfitting, acceptable
Elastic Net	0.9564	0.9090	0.0474	Mild overfitting, acceptable
Logistic Regression	0.904	0.898	0.006	Minimal overfitting
NSC	0.895	0.890	0.005	Minimal overfitting
Linear SVC	0.9580	0.8964	0.0616	Likely overfitting
PLS-DA	0.9525	0.8649	0.0875	Likely overfitting

Table 10. *Microarray Models - Overfitting Analysis*

Model	Train AUC	CV AUC	AUC Gap	Assessment
XGBoost	0.9622	0.9245	0.0377	Mild overfitting, acceptable
Random Forest	0.9839	0.9235	0.0604	Moderate overfitting
Elastic Net	0.9479	0.8988	0.0491	Mild overfitting, acceptable
Logistic Regression	0.903	0.893	0.010	Minimal overfitting
NSC	0.890	0.884	0.006	Minimal overfitting
Linear SVC	0.9492	0.8967	0.0525	Likely overfitting
PLS-DA	0.9448	0.8566	0.0882	Likely overfitting

3.5. Comparison of signal resolution

15 of the top 20 ranked RNA transcripts had no corresponding microarray probe (Table 11), and the remaining 5 transcripts were not selected by RFE despite having a corresponding microarray probe, suggesting weaker gene-level signals. The 3 highest importance RNA-Seq transcripts collectively account for over 60% of total RNA-seq model importance but carry no Microarray representation. Despite this, the RNA-seq model has a modest advantage (ROC-AUC 0.937 vs 0.919). The microarray model has better recall (0.882 VS 0.667), which is clinically important.

Table 11. *Top 20 selected features of the RNA-seq-based model and the corresponding microarray probe*

RNA transcript	RNA AUC	Microarray AUC	Difference in AUC	Number of probes targeting RNA transcript gene	Probe	Signal strength
ETV1	0.69	0.55	0.13	1	38946	Stronger in RNA-Seq
ETV4	0.73	0.59	0.14	1	42460	Stronger in RNA-Seq
GMPS	0.85	NaN	NaN	0	NaN	No Microarray probe
H2AFZ	0.84	NaN	NaN	0	NaN	No Microarray probe
HMGB3	0.85	NaN	NaN	0	NaN	No Microarray probe
IGSF9	0.78	NaN	NaN	0	NaN	No Microarray probe
JPH1	0.82	NaN	NaN	0	NaN	No Microarray probe
KIAA1244	0.58	NaN	NaN	0	NaN	No Microarray probe
KIAA1456	0.83	NaN	NaN	0	NaN	No Microarray probe

KIF18A	0.85	NaN	NaN	0	NaN	No Microarray probe
LMNB2	0.81	0.62	0.19	2	36627	Stronger in RNA-Seq
LOC100128288	0.84	NaN	NaN	0	NaN	No Microarray probe
LOC100190939	0.83	NaN	NaN	0	NaN	No Microarray probe
LOC100287482	0.74	NaN	NaN	0	NaN	No Microarray probe
MAMLD1	0.81	NaN	NaN	0	NaN	No Microarray probe
MAP6	0.78	NaN	NaN	0	NaN	No Microarray probe
MINK1	0.84	0.69	0.15	1	43985	Stronger in RNA-Seq
MIS18BP1	0.81	NaN	NaN	0	NaN	No Microarray probe
MRPL11	0.84	NaN	NaN	0	NaN	No Microarray probe
NEAT1	0.75	0.55	0.20	1	41524	Stronger in RNA-Seq

3.6. Model efficiency

We evaluated model efficiency by titering down the number of input features while assessing XGBoost performance across both platforms. The aim was to understand the minimum number of features required to achieve optimal predictive performance.

Performance plateaued at 20 features on both platforms (Figure 4), confirming that a compact gene signature captures most of the predictive signals. When we restricted the input to only 10 features, RNA-seq consistently outperformed microarray across almost all metrics (Table 12).

Table 12. Performance of models trained on 10 features

Metric	Microarray (10 Features)	RNA-Seq (10 Features)
Accuracy	0.84	0.84
Balanced Accuracy	0.80	0.86
Sensitivity (Recall)	0.75	0.88
Specificity	0.86	0.83
F1-Score	0.65	0.69
MCC	0.55	0.62
ROC-AUC	0.89	0.91

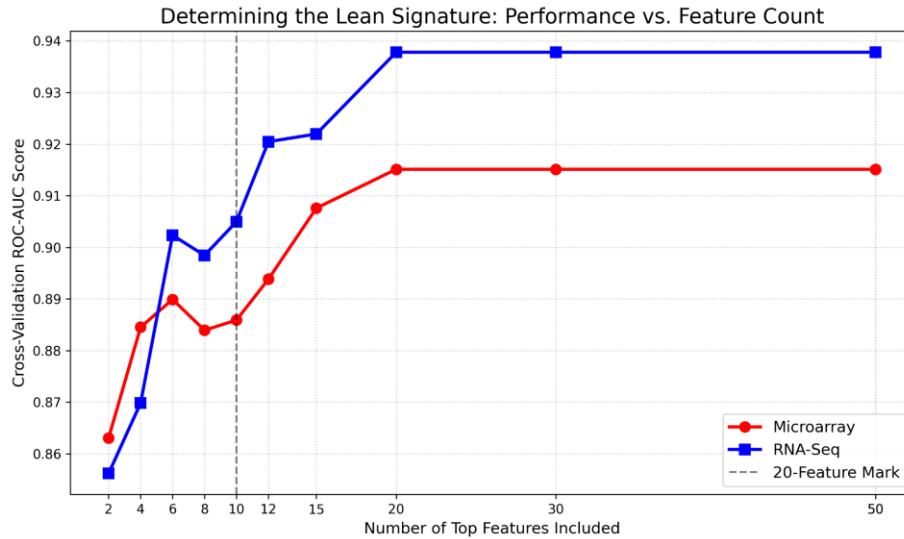


Figure 4. Estimation of the performance of the microarray and RNA-Seq-based models based on the number of features used

3.7. MYCN status and INSS stage independence of the model

Our model was MYCN status and patient age-independent. The models were able to identify unfavourable patients misclassified by the INSS staging system. The microarray model correctly identified 4 out of 7 favorable patients who died, while the RNA-seq model was only able to identify 2. Both platforms identified 57 and 58 out of 77 patients in unfavourable stages who survived, showing that gene profiling provides new survival insights that INSS staging alone misses. (Figure 5).

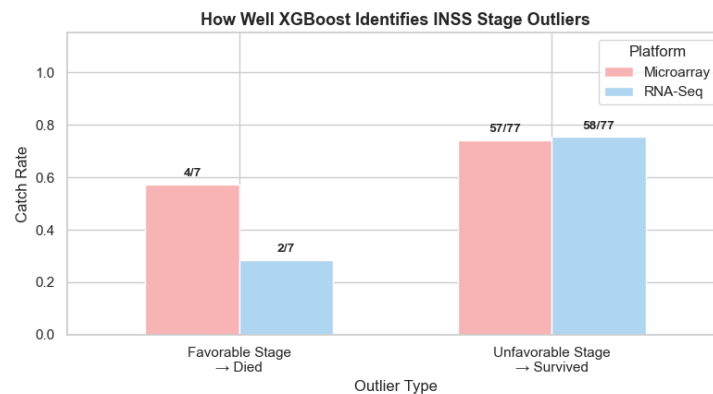


Figure 5. Identification of favorable patients (INSS stages 1, 2, 4S) that died and unfavorable patients (INSS stages 3, 4) that survived

4. Discussion

The feature selection pipeline was designed to be statistically principled and computationally scalable.

Random Forest RFE was chosen as an efficient, stable method for ranking correlated features of high-dimensional data (Bahl et al., 2019; Darst et al., 2018). All models, including linear models, were evaluated on the same selected features. This may introduce a subtle bias towards tree-based models that capture non-linear importance. Hence, there is potential to improve the performance of linear classifiers. Using a single consistent feature set across all models ensured comparability in performance differences that reflect genuine model capacity and platform information content, rather than differences in features. Another study found that models using 100–1,000 features consistently outperformed leaner signatures, arguing against aggressive feature minimisation in predictive models. This suggests that the optimal signature size may be model-dependent (Zhang et al., 2015).

Interestingly, most of RNA seq's predictive power comes from transcripts whose genes the microarray model did not have a corresponding probe. Some RFE-selected RNA features were unannotated or classified as long non-coding RNAs (E.g. LOC100128288, LOC100287482), novel transcribed regions, or neuroblastoma-specific isoforms, which are increasingly found relevant in pediatric cancers (Lee et al., 2014; Vercouillie et al., 2024; Zhang et al., 2015). Although their biological functions are still poorly characterized, lncRNAs often play important regulatory roles, suggesting that clinically relevant features exist beyond currently annotated genes. The ability of RNA-seq-based models to identify them may be valuable for future experimental validation.

XGBoost outperformed the other models on F1 and MCC (the most informative metrics that account for class imbalance). This could be due to a weighted voting mechanism and the combination of complex decision trees, making it robust to imbalanced datasets (Fan et al., 2025). Sequential gradient boosting iteratively corrects residual errors and weights misclassified minority-class samples appropriately. Linear models underperform as they assume linear decision boundaries that are unlikely to capture complex transcriptional programmes causing neuroblastoma pathogenesis.

While RNA-seq provides a broader dynamic range, improved sensitivity, and the ability to detect novel transcripts, splice variants, and non-coding RNAs compared to microarrays, it still requires greater computational infrastructure and more complex analytical workflows. Although RNA-seq remains more expensive per sample, the cost difference has narrowed. Recent estimates suggest microarrays are typically only \$50–100 cheaper per sample. In parallel, newer barcoded and 3' RNA-seq approaches have reduced sequencing costs and in some cases by up to 25-fold compared with traditional library preparation methods. Despite these advances, the performance gains observed in this study were relatively modest, suggesting that an immediate replacement of microarray platforms may not yet be justified for routine clinical deployment.

Given its competitive performance and stronger recall in our analysis, microarray may currently represent a more pragmatic clinical option, while RNA-seq remains valuable for discovery-driven feature selection and refinement of predictive gene signatures. (Antoszewski et al., 2026; “RNA-sequencing vs microarrays,” n.d.; Zhang et al., 2015)

4. 1. Limitations & Future Directions

Our analysis has several limitations. A 4:1 class imbalance reduced sensitivity despite class weighting, most notably in the RNA-seq model recall of 0.667. Model evaluation relied entirely on cross-validation since test labels were unavailable. No independent external cohort was used, limiting assessment of true generalisability across patient populations. Feature selection and parameter sweeps were both based on Random Forest models, which may bias feature selection toward non-linear patterns and disadvantage linearly separable signals. In addition, only 5 of the 20 RNA-seq features could be matched to Microarray probes, restricting direct cross-platform comparison. A small train-to-CV performance gap (~ 0.04) also suggests mild residual overfitting. The study focused primarily on predictive performance, but the biological interpretation of selected genes could be explored. Feature selection can be improved by selecting features based on their biological impact, for example, restricting features to known cancer driver pathways or applying SHAP-based interpretation. A hybrid model integrating the lean signature with clinical covariates could improve sensitivity without sacrificing specificity.

Finally, only a relatively small set of machine learning approaches was explored without more advanced ensemble or deep learning architectures.

5. Conclusion:

Both platforms produced strong prognostic classifiers using a lean 20-feature signature, with XGBoost achieving ROC-AUC above 0.91 on each. The models operate independently of MYCN status and age, and provide complementary value to INSS staging. RNA-Seq demonstrates higher efficiency and richer biomarker content, but the Microarray model's superior recall makes it more suitable for immediate clinical use.

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